

ZAK MCP Gateway v0.2

4 Oct 2026 · Build-plan upgrade

Implemented the architecture and build plan: result handles, trimming of results, safety, stats, and an honest A/B harness.

In one picture: The concierge learns to hand you a one-page summary and keep the full file in the back office (a coat-check ticket).

What's new

Feature	What it does (simply)	Where to switch it
mcp_get_result + handles	Big results stay in the gateway; page, grep, pick fields, or raw.	results.*
Noise removal + TSV tables	Drop API links and IDs; lists become tables.	results.dropKeys , tsv
Tool index in instructions	One line per server listing tool names.	discovery.catalogInInstructions
Argument + name repair	per_page → perPage, github::x → github_x; errors return the signature.	–
Safety policy	Read-only modes, allow/deny lists, writes need confirm: true, injection warning.	safety.*
Stats + report + budget	JSONL savings log, npm run report , zak://stats, session token budget.	stats.*
Remote servers + \${VAR} secrets	Streamable HTTP/SSE servers; secrets only from the environment.	url, headers, env
Passthrough baseline	--passthrough = plain MCP through the same config, for fair A/B tests.	CLI flag
Agent UI A/B	Same Gemini question through plain MCP and the gateway, real usage numbers.	agent-ui/

Fixed

Problem	Result
Cache key collision	Every call hashed to the same key
Caching never on	The read/write check was bypassed
Double-encoded results	Results were bigger than the raw reply
Placeholder token overrode the env	GitHub 401
github:: vs github__ names; per_page ignored	Tool not found; wrong page size
Agent UI made-up savings number	Replaced with real A/B usage

Measured

What	Result
Tool definitions per request (GitHub)	3,600 → 439 tokens (−88%)
7 tool results	157,173 → 7,452 tokens (−95%)
Whole 7-step agent task	805,337 → 55,130 tokens (−93%)
Live Gemini A/B, one GitHub question	20,806 → 2,522 input tokens (−87.9%)
Gateway in front of the company lean server	Lost to lean alone (243k vs 98k light task): extra searches and re-paging

Notes

- Lesson that shaped v0.3: small answers are not enough; extra turns re-send the whole conversation.

All versions: docs/versions · Full explanation: docs/ZAK_Gateway_How_It_Works.pdf